

西方经济学

Part 1 Introduction

Lecture 1B Other Measures

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Supplement Readings

西方经济学

- (1) M9; S2.¹
- (2) U.S. Bureau of Labor Statistics, U.S. Department of Commerce, FRED Economic Data.
- (3) 其他资料: [GM] THE DATA OF MACROECONOMICS



¹M 指代马工程教材, S 指代课外阅读材料沈坤荣教程。

学习目标

西方经济学

- (1) 掌握 price and quantity Indexes 的计算方法。
- (2) 掌握劳动市场常见指标。
- (3) 领会马工程教材精神。



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Outline

- 1 Measuring the Growth of GDP: Quantity Indexes
- 2 Measuring the Growth of Prices: Price Indexes
 - Examples
- 3 Measuring Joblessness
- 4 马工程教材疑难重点



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Quantity Indexes

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Let year 0 be the base year. A **Laspeyres quantity index** for year t is defined as

$$QI_{t,0}^L = \frac{\sum_i P_{i0} Q_{it}}{\sum_i P_{i0} Q_{i0}} \times 100 = \sum_i \left(\frac{P_{i0} Q_{i0}}{\sum_i P_{i0} Q_{i0}} \cdot \frac{Q_{it}}{Q_{i0}} \right) \times 100.$$

A **Paasche quantity index** for year t is defined as

$$QI_{t,0}^P = \frac{\sum_i P_{it} Q_{it}}{\sum_i P_{it} Q_{i0}} \times 100 = \sum_i \left(\frac{P_{it} Q_{i0}}{\sum_i P_{it} Q_{i0}} \cdot \frac{Q_{it}}{Q_{i0}} \right) \times 100.$$

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Laspeyres Quantity Index

3-5 国内生产总值指数

本表按不变价格计算。

(1978年=100)

年 份	国 民 总收入	国内生产 总 值	人均国内 生产总值	人均国民 总收入
1978	100.0	100.0	100.0	100.0
1979	107.6	107.6	106.2	106.2
1980	116.0	116.0	113.1	
1981	121.9	122.0	117.3	
1982	133.1	132.9	126.0	
1983	147.8	147.3	137.6	
1984	170.5	169.6	156.4	
1985	192.9	192.4	175.1	

“国内生产总值指数”指 Laspeyres quantity index。1978 年 =100 指 base year=1978。上年 =100 指 base year=t-1。 $\sum_i P_{i0} Q_{it}$ is called the constant-price real GDP.

3-4 国内生产总值指数

本表按不变价格计算。

(上年=100)

年 份	国 民 总收入	国内生产 总 值	人 均 国内生产 总值
1978	111.7	111.7	
1979	107.6	107.6	
1980	107.8	107.8	
1981	105.1	105.1	
1982	109.2	109.0	
1983	111.0	110.8	
1984	115.3	115.2	
1985	113.2	113.4	

3-3 不变价国内生产总值

单位：亿元

年 份	国内生产 总 值
	按1970年价格计算
1978	3593.0
1979	3865.8
1980	4168.6
	按1980年价格计算
1980	4587.6
1981	4822.1
1982	5257.0
1983	5823.1
1984	6707.8
1985	7608.7

109.2 109.5

113.7 113.8

111.9 111.7

Quantity Indexes: Chain-Weighted

A **Fisher quantity index** for a pair of periods, $(t-1, t)$, is defined as

$$QI_{t,t-1}^F = \sqrt{QI_{t,t-1}^L \times QI_{t,t-1}^P}$$

Let year 0 be the reference year. The **chain-type** quantity index for year t is calculated as follows.

$$QI_{t,0}^F = \frac{QI_{t,t-1}^F}{100} \times \frac{QI_{t-1,t-2}^F}{100} \times \frac{QI_{t-2,t-3}^F}{100} \times \cdots \times \frac{QI_{2,1}^F}{100} \times \frac{QI_{1,0}^F}{100} \times 100.$$



Table 1.1.3. Real Gross Domestic Product, Quantity Indexes

[Index numbers, 2012=100]

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$$Q_t^F, 2012$$

Line		2010	2011	2012	2013	2014	2015
Line							
1	Gross domestic product	96.278	97.770	100.000	101.842	104.172	106.991
2	Personal consumption expenditures	97.001	98.651	100.000	101.488	104.236	107.654
3	Goods	95.821	97.913	100.000	103.147	107.351	112.457
4	Durable goods	89.783	94.359	100.000	106.106	113.751	122.409
5	Nondurable goods	98.707	99.575	100.000	101.803	104.482	108.029
6	Services	97.581	99.015	100.000	100.684	102.748	105.400
7	Gross private domestic investment	84.542	90.097	100.000	106.855	112.871	119.071
8	Fixed investment	84.852	90.875	100.000	105.551	112.494	116.798
9	Nonresidential	84.068	91.354	100.000	104.127	111.649	114.247
10	Structures	86.098	88.465	100.000	101.256	112.390	111.408
11	Equipment	79.434	90.113	100.000	104.662	111.968	115.371
12	Intellectual property products	89.697	95.282	100.000	105.440	110.533	114.747
13	Residential	88.660	88.532	100.000	112.369	116.688	128.567
14	Change in private inventories	---	---	---	---	---	---
15	Net exports of goods and services	---	---	---	---	---	---
16	Exports	89.712	96.141	100.000	102.971	106.970	107.260
17	Goods	89.529	96.202	100.000	102.940	107.526	107.146
18	Services	90.118	96.005	100.000	103.041	105.790	107.427
19	Imports	93.126	97.610	100.000	101.213	106.436	111.956
20	Goods	92.325	97.418	100.000	102.007	107.686	113.915
21	Services	96.937	98.540	100.000	97.457	100.596	103.062
22	Government consumption expenditures and gross investment	105.469	102.113	100.000	97.585	96.708	98.466

In the NIPAs, the annual changes in quantities and prices are calculated using a Fisher formula that incorporates weights from 2 adjacent years. For example, the 2014–2015 change in real GDP uses prices for 2014 and 2015 as weights, and the 2014–2015 change in prices uses quantities for 2014 and 2015 as weights. These annual changes are "chained" (multiplied) together to form time series of quantity and price indexes.

from "Estimates for NIPA aggregates", Chapter 4, NIPA Handbook

Table 1.1.6. Real Gross Domestic Product, Chained Dollars

[Billions of chained (2012) dollars]

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$RGDP^F_{t,2012}$

Line		2010	2011	2012	2013	2014	2015
Line							
1	Gross domestic product	15,649.0	15,891.5	16,254.0	16,553.3	16,932.1	17,390.3
2	Personal consumption expenditures	10,716.0	10,898.3	11,047.4	11,211.7	11,515.3	11,892.9
3	Goods	3,485.7	3,561.8	3,637.7			
4	Durable goods	1,027.3	1,079.7	1,144.2			
5	Nondurable goods	2,461.3	2,482.9	2,493.5			
6	Services	7,230.4	7,336.7	7,409.6			
7	Gross private domestic investment	2,216.5	2,362.1	2,621.8			
8	Fixed investment	2,164.2	2,317.8	2,550.5			
9	Nonresidential	1,781.0	1,935.4	2,118.5			
10	Structures	412.8	424.1	479.4			
11	Equipment	781.2	886.2	983.4			
12	Intellectual property products	588.1	624.8	655.7			
13	Residential	383.0	382.5	432.0			
14	Change in private inventories	57.3	46.7	71.2			
15	Net exports of goods and services	-589.4	-571.0	-551.6			
16	Exports	1,989.5	2,132.1	2,217.7			
17	Goods	1,369.4	1,471.5	1,529.6			
18	Services	620.1	660.6	688.1			
19	Imports	2,578.9	2,703.1	2,769.3			
20	Goods	2,117.3	2,234.1	2,293.3			
21	Services	461.5	469.1	476.1			
22	Government consumption expenditures and gross investment	3,308.0	3,202.7	3,136.5			
23	Federal	1,348.4	1,312.0	1,287.0			
24	National defense	861.3	842.9	814.2			
25	Nondefense	487.0	469.1	472.8			
26	State and local	1,959.8	1,890.8	1,849.5	1,044.4	1,047.0	1,002.2
27	Residual	-11.3	-3.7	0.0	-0.3	0.6	10.7

The chain-weighted real GDP or the chained real GDP for year t is given by

$$RGDP^F_{t,0} = \left(\sum_i P_{i0} Q_{i0} \right) \times \frac{Q^F_{t,0}}{100}.$$

Because the formula for the chain-type quantity indexes uses weights of more than one period, the corresponding chained-dollar estimates are usually **not additive**. The residual line is the difference between the first line and the sum of the most detailed lines.

Outline

1 Measuring the Growth of GDP: Quantity Indexes

2 Measuring the Growth of Prices: Price Indexes

■ Examples

3 Measuring Joblessness

4 马工程教材疑难重点

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Price Indexes

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Let year 0 be the base year. A **Laspeyres price index** for year t is defined as

$$PI_{t,0}^L = \frac{\sum_i P_{it} Q_{i0}}{\sum_i P_{i0} Q_{i0}} \times 100 = \sum_i \left(\frac{P_{i0} Q_{i0}}{\sum_i P_{i0} Q_{i0}} \cdot \frac{P_{it}}{P_{i0}} \right) \times 100.$$

A **Paasche price index** for year t is defined as

$$PI_{t,0}^P = \frac{\sum_i P_{it} Q_{it}}{\sum_i P_{i0} Q_{it}} \times 100 = \sum_i \left(\frac{P_{i0} Q_{it}}{\sum_i P_{i0} Q_{it}} \cdot \frac{P_{it}}{P_{i0}} \right) \times 100.$$

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CPI and PPI

The **consumer price index (CPI)** is a measure of the average current level of prices of consumer goods and services purchased by households in an economy relative to the one in the reference year. The market basket for calculating CPI is called the consumption basket. CPI excluding food and energy is called **the core CPI**. The rate of inflation implied by the core CPI is called **the core rate of inflation**. (see **CPI** in BLS)

The **producer price index (PPI)** is a measure of the average current level of selling prices received by domestic producers of goods and services in an economy relative to the one in the reference year. (see **PPI** in BLS)



Price Indexes: Chain-Weighted

A **Fisher price index** for a pair of adjacent years, $(t-1, t)$, is defined as

$$PI_{t,t-1}^F = \sqrt{PI_{t,t-1}^L \times PI_{t,t-1}^P}.$$

Let year 0 be the reference year. The **chain-type** price index for year t is calculated as follows.

$$PI_{t,0}^F = \frac{PI_{t,t-1}^F}{100} \times \frac{PI_{t-1,t-2}^F}{100} \times \frac{PI_{t-2,t-3}^F}{100} \times \cdots \times \frac{PI_{2,1}^F}{100} \times \frac{PI_{1,0}^F}{100} \times 100.$$



Table 1.1.4. Price Indexes for Gross Domestic Product

[Index numbers, 2012=100]

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$$PI_{t,2012}^F$$

Line		2010	2011	2012	2013	2014	2015
Line							
1	Gross domestic product	96.164	98.157	100.000	101.769	103.662	104.662
2	Personal consumption expenditures	95.747	98.170	100.000	101.354	102.887	103.116
3	Goods	95.183	98.773	100.000	99.407	98.920	95.896
4	Durable goods	102.107	101.280	100.000	97.968	95.429	93.358
5	Nondurable goods	92.182	97.652	100.000	100.082	100.599	97.092
6	Services	96.017	97.875	100.000	102.322	104.880	106.796
7	Gross private domestic investment	97.687	98.704	100.000	100.979	102.922	103.535
8	Fixed investment	97.568	98.641	100.000	101.091	103.172	104.075
9	Nonresidential	97.416	98.559	100.000	100.251	101.469	101.909
10	Structures	92.006	95.362	100.000	101.455	107.198	109.403
11	Equipment	99.471	99.447	100.000	99.787	99.169	98.671
12	Intellectual property products	98.306	99.517	100.000	100.081	100.791	101.374
13	Residential	98.317	99.049	100.000	105.054	111.118	114.114
14	Change in private inventories	---	---	---	---	---	---
15	Net exports of goods and services	---	---	---	---	---	---
16	Exports	93.350	99.237	100.000	100.148	100.216	95.373
17	Goods	92.952	99.793	100.000	99.320	98.312	91.323
18	Services	94.230	98.001	100.000	101.987	104.466	104.431
19	Imports	92.655	99.716	100.000	98.697	97.961	90.144
20	Goods	91.960	99.868	100.000	98.059	96.752	87.644
21	Services	95.891	98.987	100.000	101.841	104.018	102.981
22	Government consumption expenditures and gross investment	95.391	98.289	100.000	102.363	104.470	104.638

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How to deflate a nominal variable

Nominal variables are affected by the change in both quantity and price. Filtering out the effect of inflation, we obtain deflated or real variable.

$$\text{GDP Deflator}_{t,0} = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_{t,0}} \times 100.$$

Real GDP can be constant-price GDP or chain-weighted real GDP.

$$\text{Deflated Variable}_{t,0} = \frac{\text{Nominal Variable}_t}{\text{Appropriate Price Index}_{t,0}/100}.$$



Personal consumption expenditures (PCE) deflator is a measure of the overall level of prices that shows the cost of the currently consumed basket of goods relative to the cost of that basket in a base year.

$$\text{PCE Deflator}_{t,0} = \frac{\text{Nominal Personal Consumption Expenditures}_t}{\text{Real Personal Consumption Expenditures}_{t,0}} \times 100.$$

- (1) Like the CPI, the PCE deflator includes only the prices of goods and services that consumers buy; it excludes the prices of goods and services that are part of investment spending and government purchases.
- (2) Also like the CPI, the PCE deflator includes the prices of imported goods.
- (3) But like the GDP deflator, the PCE deflator allows the basket of goods to change over time as the composition of consumer spending changes.



Inflation Rates

An **inflation rate** (π) is the growth rate of a price index given the reference year 0.

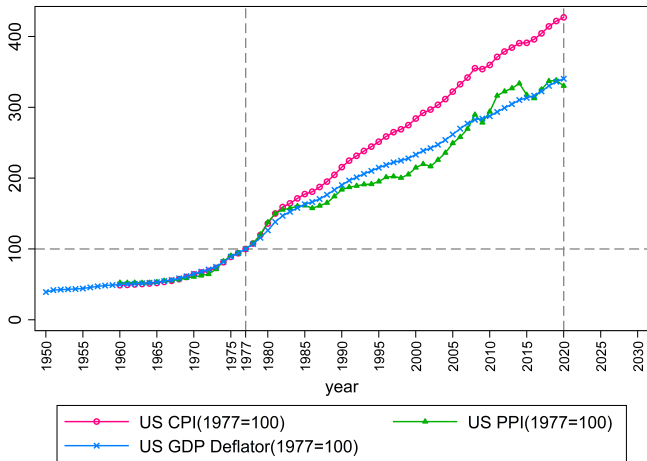
$$\pi_t = \left(\frac{\text{Price Index}_{t,0}}{\text{Price Index}_{t-1,0}} - 1 \right) \times 100\%$$

The price index can be GDP deflator, CPI, or PPI. Which one you choose depends on the purpose. The growth rates of GDP deflator, CPI and PPI measure the average price changes from the perspective of overall economy, the consumer, and the seller, respectively. Sellers' and purchasers' prices may differ due to government subsidies, sales and excise taxes, and **distribution costs**.



Example 1

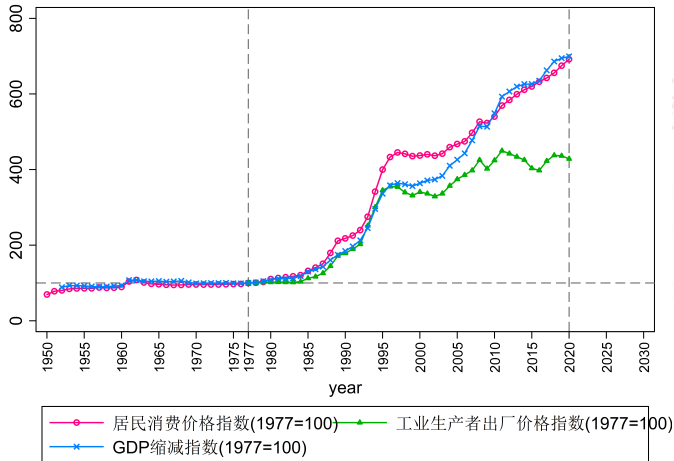
Does CPI overstate the cost of living? Does GDP deflator understate it? Why?



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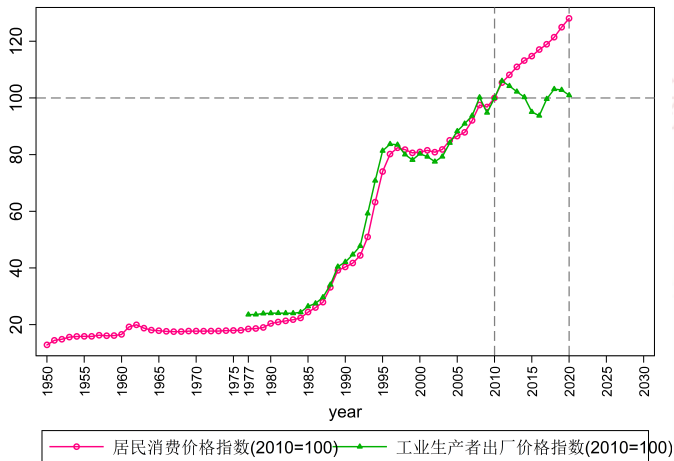
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- 1 Measuring the Growth of GDP: Quantity Indexes
- 2 Measuring the Growth of Prices: Price Indexes
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Measuring Joblessness

Employment (E) is the number of people who have a job.

Unemployment (U) is the number of people who do not have a job but are looking for one. The **labor force** (L) is the sum of employment and unemployment.

$$L = E + U$$

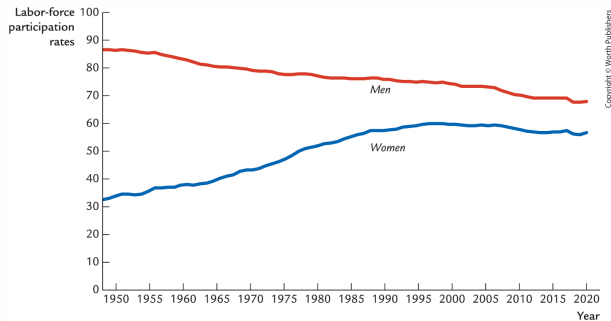
$$\text{Unemployment Rate}(u) = \frac{\text{Unemployment}(U)}{\text{Labor Force}(L)} \times 100\%$$

$$\text{Participation Rate} = \frac{\text{Labor Force}(L)}{\text{Total Working Age Population}} \times 100\%$$

$$\text{Employment/Population ratio} = \frac{\text{Employment}(E)}{\text{Total Working Age Population}} \times 100\%$$



Trends in Labor-Force Participation



Just after World War II, men and women had very different economic roles. Only 33 percent of women were working or looking for work, compared with 87 percent of men. Since then, the difference between the participation rates of men and women has gradually diminished, as growing numbers of women have entered the labor force and some men have left it. Data for 2022 show that almost 57 of women were in the labor force, compared with 68 percent of men. As measured by labor-force participation, men and women are now playing more equal roles in the economy.



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- (1) Nominal GDP 和 Real GDP 的区别是什么？为什么要区分它们？
- (2) Quantity Indexes(Laspeyres, Paasche, and Chain-type) 怎么计算？
- (3) Price Indexes(Laspeyres, Paasche, and Chain-type) 怎么计算？
- (4) 了解常见的价格指数：GDP deflator, PCE deflator, CPI, PPI。衡量生活成本时，GDP deflator and CPI 哪个会高估生活成本？
- (5) 如何计算通货膨胀率？
- (6) 掌握 Employment, unemployment, the labor force, unemployment rate, participation rate, employment/population ratio 的概念和计算。



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1 (E2, p.28)

根据马工程教材观点，应当如何评析西方经济学的价格水平指标？

2 (E2, p.28)

根据马工程教材观点，应当如何评析西方经济学的失业指标？



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